

PART 2

Connecting AI to Your Own Data

Why a model doesn't know your business · RAG · Embeddings · Chunking · Getting messy data ready

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The model doesn't know your business



"How many holiday days can I carry over into next year?"



A general model gives you a plausible, confident, **wrong** answer.

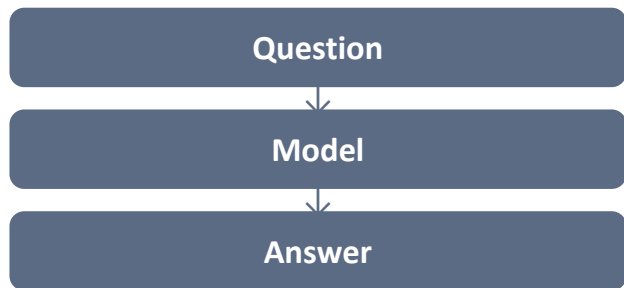
Three things a model cannot know:

- Your internal policies and procedures
- Anything that changed after its training cut-off
- Anything private to your company

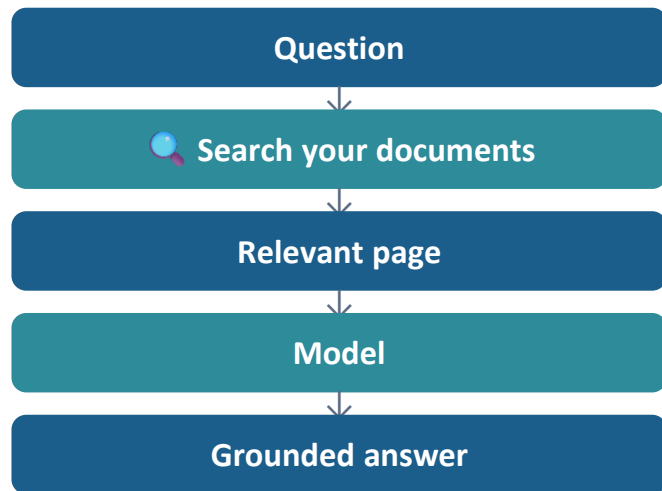
The fix isn't a bigger model. It's handing it the right document at the right moment.

RAG: giving the model an open book

Without RAG — closed book



With RAG — open book

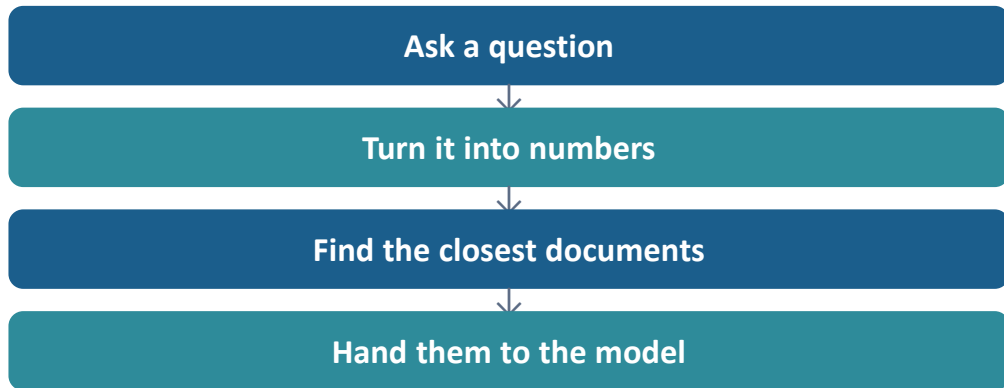


The system finds the actual HR policy page first, then answers from it — and can tell you which document it used. When the policy changes, you update the document. You don't retrain anything.

Embeddings: searching by meaning

An embedding turns a piece of text into a list of numbers that captures its meaning. Similar meanings end up close together.

-  Keyword search for **"time off"** misses a policy that says **"annual leave"**.
The words don't match, so it finds nothing.
-  Meaning-based search finds it — **"time off"** and **"annual leave"** sit right next to each other.



Chunking: how you slice matters

You can't store an 80-page handbook as one block. You cut it into chunks.

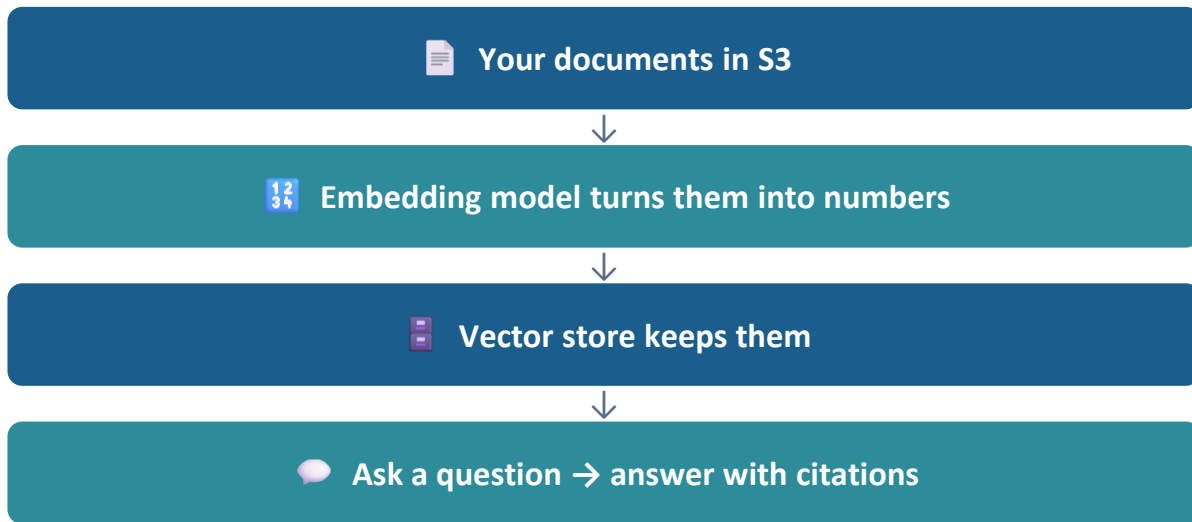
Method	What it does
Fixed size	Cut every ~300 words. Simple and cheap.
Hierarchical	Search small pieces, hand over the bigger section around it.
Semantic	Let a model decide where the topic changes. Costs more.
None	You've already split the file yourself.



Cut in the wrong place and the model sees:

"employees may carry over holiday" — without "...up to a maximum of five days."

Bedrock Knowledge Bases do this for you



 Your documents don't have to be in S3 — a knowledge base can also pull from SharePoint, Confluence, Salesforce, or crawl a website. Which is usually where a company's real knowledge actually lives.

Getting messy data ready

Service	What it does
Amazon Transcribe	Audio → text
Amazon Comprehend	Finds names, places and personal data
Bedrock Data Automation	Pulls fields out of PDFs, images, video
AWS Glue	Cleans and moves data in bulk



Practical example

10,000 recorded support calls → Transcribe writes them up → Comprehend strips out customer names → the result is a searchable knowledge base of what customers actually complain about. Nobody could read through that manually.

Key takeaways

- 1 A model knows nothing about your business until you show it
- 2 RAG = search your documents first, then answer
- 3 Embeddings search by meaning, not by keywords
- 4 Chunking decides what the model actually sees
- 5 When something changes, update a document — not a model

Next · Part 3: keeping it safe, accurate and affordable